

Solution for 2025 SAJC H2 Physics Prelim Paper 3

- 1 (a)** constant velocity in one direction [1]
constant acceleration in the perpendicular direction to the velocity [1]
- (b) (i)** Work done by spring = Loss in elastic potential energy
 $F_{\text{ave}} \times (8.0 \times 10^{-2}) = 0.24$
 $F_{\text{ave}} = 3.0 \text{ N}$ [1]
- (ii)** Using Principle of conservation of energy,
Loss in Elastic PE = Gain in GPE + Gain in KE
 $0.24 = mgh + \frac{1}{2}mv^2$ [1]
 $0.24 = (4.5 \times 10^{-2})(9.81)(8.0 \times 10^{-2} \times \sin 15^\circ) + \frac{1}{2}(4.5 \times 10^{-2})v^2$
 $v = 3.2 \text{ m s}^{-1}$ [1]
- (iii)** Using $v^2 = u^2 + 2as$,
 $v^2 = 2.4^2 + 2(-9.81 \sin 15^\circ)(0.20)$ [1]
 $v = 2.18 = 2.2 \text{ m s}^{-1}$
- (c) (i)** Speed at maximum height = $2.18 \cos 15^\circ = 2.1 \text{ m s}^{-1}$ [1]
- (ii)** Using $s_y = u_y t + \frac{1}{2} a_y t^2$, taking upward as positive,
 $-0.40 = (2.18 \sin 15^\circ)t + \frac{1}{2}(-9.81)t^2$ [1]
 $t = 0.349 = 0.35 \text{ s}$ [1]
- (iii)** Horizontal displacement = $(2.18 \cos 15^\circ)(0.349) = 0.735 \text{ m}$ [1]

